Quantitative Screening and Physical-Norm Certificates for Joint Critical Ashkin–Teller Currents

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Technical Report Assisted by AI Models
www.integralresearch.net
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Technical Report TR-Ram-OpenAI-722-M446

October 9, 2026

Abstract

The source manuscript claims joint convergence of the height and both complete current-cluster collections along the fixed critical Ashkin–Teller line. We examine its covariance, screening, conditional-field and local-kernel interfaces without equating a finite covariance calculation with that global claim. The principal local refinement replaces the qualitative analytic translation argument by an explicit slit-ellipse interpolation exponent. It yields a finite error recursion through rigid moves, terminal-color truncation and shielding, with every required uniform bound exposed. A positive Cauchy-mixture defect controls both bounded kernel tests and a reflected Laplace insertion at positive clearance. Exact Gaussian-shift variation improves fixed-buffer net constants, while compatible weak coordinates preserve the physical Dirichlet norm and its residual obligations. Worked rational certificates illustrate these estimates. The results do not supply the missing numerical constants, a mesh convergence rate, or a new proof of the all-generation joint local-set identification. Contribution assessment: C2/5, the report supplies explicit conditional screening and local-kernel error budgets, but their classical ingredients and unevaluated upstream moduli make the demonstrated increment modest.

1 Object, normalization and scope

The object under examination is the September 25, 2026 source *The joint scaling limit of critical Ashkin–Teller currents* [1]. Its claimed conclusion is substantially stronger than height covariance convergence. Throughout this report the coupling parameter is fixed:

$$J > 0, \quad U \le J, \quad \sinh(2J) = e^{-2U}, \qquad t = (1 + e^{4U})^{-1/2} \in [1/2, 1), \quad c = t^{-1}.$$

Set

$$\vartheta = \arcsin(c/2), \quad g = \frac{8\vartheta}{\pi}, \quad \lambda = \frac{\pi}{2}, \quad a = \lambda\sqrt{g}, \quad a^2 = 2\pi\vartheta. \tag{1}$$

All constants depending on the parameter are allowed to deteriorate as $t \uparrow 1$. No uniform assertion at that excluded endpoint is made. The four-state Potts endpoint $t = 1/2$ is included by the source.

Let D be a bounded Jordan domain with the source's admissible simple polygonal lattice approximations, including boundary parametrization convergence and compact containment. Primal spins have wired + boundary conditions; the weak dual has free boundary conditions. On an edge where both spin pairs agree, the *joint* primal/dual current occupation table is

$$\mathbb{P}((1,0)) = 1 - t, \qquad \mathbb{P}((0,1)) = 1 - t, \qquad \mathbb{P}((0,0)) = 2t - 1. \tag{2}$$

On a disagreement the allowed opposite current is forced. Thus the two currents do not cross, but they are not complements for $t > 1/2$. The auxiliary variable W used in a one-parity spatial representation is the complement of the chosen current N, not the other physical current.

Conditioning on decorations that the representation has already summed out can destroy its exact spatial factorization.

The corner height satisfies $H(f) - H(v) = \tau(v)\tau^{\dagger}(f)/2$ and has zero boundary height; the unit-increment field is $K = 2H$. Let h denote the zero-Dirichlet GFF with

$$\mathrm{Cov}(h(f), h(q)) = G_D(f, q), \qquad G_D = 2\pi(-\Delta_D)^{-1}, \qquad \|u\|_{\nabla}^2 = \frac{1}{2\pi}\int_D |\nabla u|^2. \tag{3}$$

The source claims

$$K_\delta \Longrightarrow h/a, \qquad H_\delta \Longrightarrow h/(2a) = h/(\pi\sqrt{g}) \quad \text{in } H^s_{\mathrm{loc}},\ s < -1.$$

Consequently the covariance operators of the limiting K and H are $\vartheta^{-1}(-\Delta_D)^{-1}$ and $(4\vartheta)^{-1}(-\Delta_D)^{-1}$, respectively. The primary and dual atomic smearings have the same limiting normalization; averaging the two is not summing two copies of the field.

The distinguished wired cluster is claimed to be $A_{-a,a}(h)$. The free recursion first exposes a two-valued set at levels $\pm 2\lambda$, and then makes a sign-dependent split in each hole: the positive incoming 2λ branch uses relative levels $-2\lambda, 2a - 2\lambda$, with the reflected prescription on the negative branch. All generations are retained, and both current families are constructed from the *same* h. The initial CLE carpet itself is not the retained free cluster.

The cluster topology is Hausdorff matching of singleton-augmented compact families. It ignores multiplicities and need not admit injective matching. Ordinary Hausdorff convergence separately marks the wired boundary cluster. Area measures, magnetization fields, parametrized wall curves and metric exponents are not part of this source statement.

Logical separation. Statements explicitly called source inputs below are retained hypotheses, not conclusions of the local estimates. The report proves the analytic and probabilistic inequalities it uses for its refinements. It does not independently validate every spatial estimate, external local-set theorem, or all-generation matching argument in [1]. The source is treated as a manuscript claiming the global theorem, not as a substitute for its proof or for independent acceptance of that theorem.

2 Source mechanism and the local point of attachment

The source's ten scientific modules form the following dependency chain. This is a map of the whole architecture, not a claim that every mapped proof has been read or independently verified.

Module	Mathematical role and retained obligation
01–02	Critical line, exact coupled law, height scale, canonical recursion, and noninjective compact-family topology.
03	Exact cuts with retained exterior connections; monotone comparison, circuits, terminal counts, multiplicative rare-pattern comparison and full-spin contiguity.
04	Positive transfer in four reflection frames; Cauchy disintegration, summable angular defect, axis endcaps, plane Green covariance and reflected Laplace nullity.
05	Pattern-uniform moment bounds, analytic translations, core colors, selected-arm shielding and transfer from plane to flat domains.
06	Boundary traces, parity equivalence, exclusion of diagonal contact measures, harmonic collision coefficients, Wick recursion and only then free-energy identification of the GFF multiplier.
07	Vanishing cut neighborhoods, joint retained stopped state and full conditional field across cuts, including pinched precursors.
08	Whole-cable reverse-buffer estimate, a common conditional local kernel, canonical forcing and no crossing, with separate conditioning laws.
09	Visible partition, invisible contributors, discovered mean, all positive diameters and generations, then the same-field joint limit.
10	Explicit input conventions for increment paths, slope/free energy, Dirichlet energy and conformal growth.

In module 04 the spectral covariance is a positive mixture of Cauchy normal-frequency laws. A fourth angular harmonic has nonnegative Cauchy average

$$D(k,s) = \left(\frac{|k|-s}{|k|+s}\right)^2.$$

Its sign reverses between frames separated by 45 degrees on the *same* physical annulus. This is not an assumption of 45-degree invariance of every decorated current. Summability of the defect forces rescaled mass onto $s = |k|$, but endcap and alias estimates are separately needed. The resulting Green covariance has a subsequential coefficient. Module 06 identifies Gaussian correlations at every order before using the external free-energy correspondence to fix that coefficient.

The target in module 05 is $X_\delta = K_\delta(\Delta\phi)$, with a smooth test ϕ in an interior buffer. Reflection positivity gives small $\rho_\delta = \|X_\delta\|_\Theta^2$ against a reflected insertion, not small ordinary variance. Separated complete (s, W) patterns can be very rare. Their multiplicative comparison and normalized reflection bound are essential: an additive total-variation error cannot be divided by an arbitrarily small pattern probability.

The source then translates groups of patterns rigidly. A positive normal transfer operator and a fixed backoff give a holomorphic correlation in a common half-plane, uniformly bounded after the rare-pattern normalization. Smallness for an interval of final translated layouts is returned to the original layout by a qualitative Montel/identity-theorem argument. Theorem 3.1 gives an explicit replacement for precisely this step. The required bound must be uniform in complete patterns, remaining layout parameters and mesh; a separate bound for each pattern is insufficient.

3 A quantitative analytic-translation certificate

Theorem 3.1 (Slit-ellipse interpolation)**.** *Let $c > d > 0$, $A > c$, and let $\mathcal{E}_A$ be the closed ellipse centered at c with foci $c-d, c+d$ and horizontal semiaxis A. Suppose F is holomorphic on a neighborhood of this ellipse, $|F| \le M$ there, and $|F| \le \epsilon$ on $I = [c-d, c+d]$, where $0 \le \epsilon \le M$. Then*

$$|F(0)| \le M^{1-\theta}\epsilon^\theta, \qquad \theta = 1 - \frac{\operatorname{arcosh}(c/d)}{\operatorname{arcosh}(A/d)} \in (0,1). \tag{4}$$

In particular, if F is bounded by M on $\{\Re z > -b\}$, $b > 0$, one may take $A = c + b/2$.

Proof. Off I, put

$$G(z) = \log \left| \zeta + \sqrt{\zeta^2 - 1} \right|, \qquad \zeta = (z - c)/d,$$

where the branch is chosen so that the modulus exceeds one. Equivalently G is the Green function of the complement of the segment; its modulus definition has no branch ambiguity. It is harmonic off I, continuous and zero on I. On $\partial\mathcal{E}_A$ it equals $L = \operatorname{arcosh}(A/d)$, whereas $G(0) = \operatorname{arcosh}(c/d)$. For $0 < \epsilon < M$, the harmonic function

$$\log M + (1 - G(z)/L) \log(\epsilon/M)$$

majorizes the subharmonic function $\log |F(z)|$ on the boundary of the slit ellipse. The maximum principle, with a limiting argument at zeros of F and at the two slit endpoints, gives the inequality throughout the slit domain. Evaluate at zero. Cases $\epsilon = M$ and $M = 0$ are immediate; $\epsilon = 0$ follows by the identity theorem or by decreasing positive upper bounds. The ellipse has left endpoint $-b/2$ in the half-plane case, so its closure has strictly positive clearance from the boundary of the holomorphy domain. □

This is a classical two-constants argument with explicit geometry, not a new principle of analytic continuation. Its source-specific use is that the reflection-transfer backoff supplies the half-plane and a bound uniform in rare complete patterns. Without that bound, interpolation from finitely many small real values is unstable.

Corollary 3.2 (Grid and finite-move budget)**.** *In the half-plane case, suppose every $t \in I$ has a sampled real $t' \geq 0$ with $|t - t'| \leq \tau$ and $|F(t')| \leq \epsilon_{\text{grid}}$. Then (4) holds with*

$$\epsilon_{\text{eff}} = \min\{M, \epsilon_{\text{grid}} + 2M\tau/b\}.$$

For a finite sequence of L inverse translations, suppose their half-plane bounds and interval estimates hold uniformly in all remaining parameters. With one common $M > 0$, their normalized errors obey

$$x_{j-1} \leq \min\{1, x_j + r_j\}^{\theta_j}, \qquad r_j = 2\tau_j/b_j, \quad x_j = e_j/M. \tag{5}$$

Without rounding,

$$e_0 \leq M^{1-\Theta} e_L^{\Theta}, \qquad \Theta = \prod_{j=1}^{L} \theta_j. \tag{6}$$

In particular a uniform final reflection estimate $e_L \leq \min\{M, C\sqrt{\rho_\delta}\}$ yields this same bound for the original gapped-pattern error.

Proof. For each real $x \geq 0$, the disk of radius $b/2$ about x lies in the half-plane. Cauchy's derivative estimate gives $|F'(x)| \leq 2M/b$. Integrating on the segment between t, t' proves the first claim. Apply theorem 3.1 one move at a time. Uniformity in the other parameters is exactly what permits its iteration. Repeated exponentiation gives (6); reflection Cauchy–Schwarz supplies the last stated input, not the present theorem. □

For fixed macroscopic layouts the source's moves and positive backoffs are finite, and the parity-preserving lattice step is $O(\delta)$. Nevertheless Θ can be small, and M can depend strongly on the geometry and parameter. The corollary neither chooses a mesh-dependent number of moves nor exchanges shrinking-buffer and mesh limits.

3.1 An exact rational ellipse example

Take $c = 5/4$, $d = 1$, $b = 7/4$, so $A = 17/8$. Then $\operatorname{arcosh}(c) = \log 2$, $\operatorname{arcosh}(A) = \log 4$, and $\theta = 1/2$. For the Chebyshev polynomial T_m, put

$$F_m(z) = \frac{T_m(z - 5/4)}{T_m(17/8)}, \qquad \epsilon_m = \frac{1}{T_m(17/8)}.$$

On the ellipse $|F_m| \le 1$ and on the slit $|F_m| \le \epsilon_m$. Indeed $T_m((w+w^{-1})/2) = (w^m + w^{-m})/2$ and the ellipse corresponds to $|w| = 4$. At zero the claimed inequality reduces to

$$T_m(5/4)^2 \le T_m(17/8) = 2T_m(5/4)^2 - 1.$$

This is exact rational arithmetic. These polynomials test the *ellipse* version, not boundedness on the entire half-plane. Their exponential decay rates also explain why a fractional power of the supplied smallness is natural. They are not current-model samples.

4 From the angular defect to a reflected insertion

Proposition 4.1 (Cauchy-mixture comparison)**.** *For $b, s > 0$ let $q_s(\ell) = s/[\pi(s^2 + \ell^2)]$ and $D_b(s) = ((s-b)/(s+b))^2$. In the convention $\mathrm{TV}(P,Q) = \sup_E |P(E) - Q(E)|$,*

$$\mathrm{TV}(q_s, q_b) = \frac{2}{\pi} \arcsin \frac{|s-b|}{s+b} \le \sqrt{D_b(s)}. \tag{7}$$

The same upper bound holds after periodization to any common circle, including the limits $s = 0$ and $s = \infty$. If $\nu(\mathrm{d}k, \mathrm{d}s)$ is a finite positive measure with mass M_ν on $0 < b_0 \le |k| \le B$, and $\Delta_\nu = \int D_{|k|}(s)\,\mathrm{d}\nu$, replacing each normal law q_s by $q_{|k|}$ changes every bounded joint test $F(k,\ell)$ by at most

$$2\|F\|_\infty \sqrt{M_\nu \Delta_\nu}. \tag{8}$$

Proof. For $s > b$, the densities cross at $|\ell| = \sqrt{sb}$. Their central probability difference is $\frac{2}{\pi}[\arctan\sqrt{s/b} - \arctan\sqrt{b/s}]$. This equals the expression in (7); exchange s, b for the other case. The inequality follows from $\arcsin x \le \pi x/2$ on $[0,1]$. Pushforward under reduction modulo the period contracts total variation. The endpoint upper bound is simply $\mathrm{TV} \le 1$; the periodized infinite-scale law is uniform. Integrate the bounded-test estimate $2\|F\|_\infty\sqrt{D}$ and apply Cauchy–Schwarz to the positive mixing measure. □

The measure in this proposition need not have a Lebesgue tangential marginal. Therefore a small defect alone does not fix either the radial Green measure or its amplitude. The four-frame and endcap steps remain.

Theorem 4.2 (Positive-clearance Laplace certificate)**.** *Let ν be as above, with finite $s \ge 0$, symmetric under $k \mapsto -k$. Define a covariance by*

$$C_\nu((x,y),(x',y')) = \int e^{ik(x-x')} e^{-s|y-y'|} \nu(\mathrm{d}k, \mathrm{d}s).$$

For real $\phi \in C_c^\infty(\mathbb{R} \times (h,\infty))$, $h > 0$, let $\Theta(x,y) = (x,-y)$ and $\widehat{\phi}(k,y) = \int e^{-ikx}\phi(x,y)\,\mathrm{d}x$. The reflected quadratic form of the insertion $\Delta\phi$ is

$$\mathcal{R}_\nu(\Delta\phi) = \int (s^2 - k^2)^2 \left| \int e^{-sy} \widehat{\phi}(k,y)\,\mathrm{d}y \right|^2 \nu(\mathrm{d}k, \mathrm{d}s), \tag{9}$$

$$\le L_\phi^2 C(h,B) \Delta_\nu, \qquad L_\phi = \sup_{b_0 \le |k| \le B} \int |\widehat{\phi}(k,y)|\,\mathrm{d}y \le \|\phi\|_1, \tag{10}$$

where

$$C(h,B) = \begin{cases} B^4, & B \ge 2/h, \\ (2/h)^4 e^{-4+2hB}, & 0 < B < 2/h. \end{cases}$$

Proof. Reflected points lie on opposite sides, so the normal kernel is $e^{-s(y+y')}$ for $y, y' > h$. Fourier transformation in x and two integrations by parts in y convert each Laplacian to $s^2 - k^2$. Compact support away from the reflection line eliminates all boundary terms. Fubini is justified by finite mass, bounded tangential frequency and the exponential normal damping, even when s is unbounded. This proves (9). Now $(s^2-k^2)^2 = D_{|k|}(s)(s+|k|)^4$ and the squared Laplace transform is bounded by $L_\phi^2 e^{-2hs}$. Maximizing $(s+B)^4 e^{-2hs}$ over $s \ge 0$ gives the displayed two cases. □

This is a continuum finite-band statement. To apply it quantitatively to ρ_δ one must separately certify, for example,

$$\rho_\delta \le L_\phi^2 C(h,B)\Delta_{\nu_\delta} + \xi_\delta, \qquad \xi_\delta \ge 0. \tag{11}$$

Here ξ_δ includes out-of-band mass, aliases, periodization and the physical test/interpolation discrepancy. The source controls their ordered limiting behavior but this report does not assign them numerical mesh rates. Dropping ξ_δ, treating a signed approximate measure as positive, or letting h vanish without tracking $C(h,B)$ invalidates the certificate.

4.1 Contact variance is not a defect

For the normalized limiting field K, integration by parts gives

$$\mathrm{Cov}(K(\Delta\phi), K(\Delta\psi)) = \vartheta^{-1}\int_D \nabla\phi\cdot\nabla\psi \quad (\phi,\psi\in C_c^\infty(D)). \tag{12}$$

Thus the ordinary variance is generally positive. Reflection-nullity is consistent with it because reflected and original gradients have disjoint supports. For an explicit Sobolev fixture take $\phi(z) = (1-|z-(0,2)|^2)_+^2$ in a sufficiently large domain. Direct radial integration gives $\int|\nabla\phi|^2 = 4\pi/3$. The function is not C_c^∞; (12) applies by H_0^1 approximation, since the pairings extend in that norm. At $U=0$, $\vartheta=\pi/4$, so its K variance is $16/3$ and its H variance is $4/3$, while its reflected covariance is zero. This fixture detects a factor-four height error and the incorrect identification of reflected norm with ordinary variance.

5 A finite screening error ledger

Proposition 5.1 (Terminal-color optimization)**.** *Let $e, K, A_0 \ge 0$. Assume the exact color decomposition of source module 05: sealed colors are independent and fair, and on the event that at most m unsealed terminals are present, the sum of the absolute unnormalized conditional expectations is at most $2^m e$. Suppose, under the original law,*

$$\mathbb{E}|X|^2 \le K^2, \qquad \mathbb{P}(N>m) \le A_0 e^{-\kappa m} \quad (m=0,1,\dots), \quad \kappa>0.$$

Writing $B_0 = K\sqrt{A_0}$, the core conditional mean satisfies

$$\|\mathbb{E}[X \mid \textit{patterns and core colors}]\|_1 \le \inf_{m\ge 0,\ m\in\mathbb{Z}} \{2^m e + B_0 e^{-\kappa m/2}\}. \tag{13}$$

If $0 < e \le B_0$, this is at most

$$(1+e^{\kappa/2}) B_0^{1-\gamma} e^{\gamma}, \qquad \gamma = \frac{\kappa}{\kappa + 2\log 2}. \tag{14}$$

If $e=0$, the right side of (13) tends to zero.

Proof. For a fixed retained pattern there are at most 2^m terminal sign assignments. They need not have uniformly positive probabilities and some can be forbidden. Sum absolute *unnormalized* expectations, integrating sealed colors with total weight one. The assumed retained bound follows without division by assignment probabilities. On $\{N>m\}$, conditional expectation contraction followed by Cauchy–Schwarz gives $K\sqrt{\mathbb{P}(N>m)}$. This proves (13). Take

$$m = \left\lfloor \frac{\log(B_0/e)}{\log 2 + \kappa/2} \right\rfloor.$$

The first term is at most $B_0^{1-\gamma}e^{\gamma}$ and the second at most $e^{\kappa/2}B_0^{1-\gamma}e^{\gamma}$. □

The terminal tail is required under the original, unconditioned pattern law. We do not replace it by a conditional tail uniform in every rare complete pattern. That stronger and unnecessary premise is absent from the source. The source already has the two terms in (13); the explicit optimized power and its incorporation with (5) are the present local extraction.

Theorem 5.2 (Screening with named finite errors)**.** *Fix the source's separated layout and original plane law P, its artificial bridge law Q, and their exact separator factorization. Let $\mathscr{E}$ be the full exterior data and Y the retained core data. Suppose the two couplings of insertion/data used in that factorization have mismatch probabilities at most ε_c, and the unsuccessful-pattern event under Q has probability at most ζ. Write $r(T) = \mathbb{E}_P|X - \operatorname{clip}_T X|$, $T > 0$, and let e_{core} be any bound from proposition 5.1. Then*

$$\|\mathbb{E}_P[X \mid \mathscr{E}]\|_1 \le e_{\mathrm{core}} + 2r(T) + 4T\varepsilon_c + 2T\zeta. \tag{15}$$

With the source's selected-arm estimates $\varepsilon_c \le 4\eta(q)$ and $\zeta \le 4(1 - p(q))^M$, this becomes

$$e_{\mathrm{core}} + 2r(T) + 16T\eta(q) + 8T(1 - p(q))^M. \tag{16}$$

Here the trial count M is unrelated to the holomorphic upper bound in theorem 3.1. At each fixed layout, corollary 3.2 supplies e in proposition 5.1 whenever its uniform analytic premises hold.

Proof. For bounded Z, the norm $\|\mathbb{E}[Z \mid \mathscr{A}]\|_1$ is the supremum of $\mathbb{E}[ZV]$ over $\mathscr{A}$-measurable $|V| \le 1$. If a coupling agrees on both Z and the data outside an event of probability ε_c, and $|Z| \le T$, the expectations differ by at most $2T\varepsilon_c$. Apply this first to the exterior conditional norm under P and Q. Under Q, the exact separator makes the exterior conditionally independent of Z given Y on success; the exceptional histories cost at most $2T\zeta$. Apply the second coupling to return the core norm to P. These three comparisons give

$$\|\mathbb{E}_P[X^{(T)} \mid \mathscr{E}]\|_1 \le \|\mathbb{E}_P[X^{(T)} \mid Y]\|_1 + 4T\varepsilon_c + 2T\zeta.$$

Replacing the two clipped insertions by X costs $r(T)$ on each side. All integrability is under P; no moment bound under the artificially pinned law Q has been introduced. □

For instance, if $\mathbb{E}_P|X|^p \le K_p$ for some $p > 1$, then $r(T) \le K_p T^{1-p}$. The sharper scalar bound

$$r(T) \le K_p \frac{(p-1)^{p-1}}{p^p T^{p-1}}$$

follows by maximizing $(x - T)_+/x^p$ at $x = pT/(p-1)$. Either is a valid input. The resulting choice of T still has to precede the geometric choices controlling $\eta(q)$ and $p(q)$; the constants of the analytic moves depend on those choices.

Corollary 5.3 (A flat-law transfer with a supplied modulus)**.** *Fix $p > 1$. Suppose the same ring kernel g is defined under a plane law P and a flat law μ, $\mathbb{E}_P|g| \le e_P$, $\mathbb{E}_\mu|g|^p \le K_p$, and for all ring events $\mu(E) \le \omega(P(E))$, where ω is a nondecreasing contiguity bound on $[0, 1]$. Then, for every $h_0 > 0$,*

$$\mathbb{E}_\mu|g| \le h_0 + K_p^{1/p}\omega(\min\{1, e_P/h_0\})^{1-1/p}. \tag{17}$$

Proof. Markov gives $P(|g| > h_0) \le e_P/h_0$. Split at h_0, use the contiguity bound and apply Hölder to the tail under μ. □

Source module 03 supplies a uniform contiguity modulus through finite terminal counts and witness probabilities, but not an evaluated power-law modulus. Its displayed estimate has the form $\mu(E) \le \epsilon/2 + D(\epsilon)P(E)$, with $D(\epsilon)$ depending on the short-witness lower bound. One may take the infimum of these bounds as ω after capping at one. This does not justify assigning a polynomial mesh rate to (17).

5.1 A worked composition and its costs

Consider a hypothetical certified two-move layout with the rational ellipse parameters of section 3.1, common bound one, zero rounding, and final reflected error at most $\sqrt{\rho}$. Then $\Theta = 1/4$ and $e \le \rho^{1/8}$. If the terminal tail satisfies $B_0 = 1$, $\kappa = \log 4$, then

$$e_{\mathrm{core}} \le 3\sqrt{e} \le 3\rho^{1/16}.$$

The constants are deliberately kept: knowing $\rho = 2^{-64}$ yields only $e \le 2^{-8}$ and the optimized integer choice $m = 4$ gives $2^m e + 2^{-m} = 1/8$, rather than a spectacular error. If also $\mathbb{E}X^2 \le 1$, the sharp scalar clipping bound gives $2r(T) \le 1/(2T)$. With $T = 8$, $\eta(q) \le 2^{-14}$ and $(1 - p(q))^M \le 2^{-13}$, (16) is bounded by

$$\frac{1}{8} + \frac{1}{16} + \frac{1}{128} + \frac{1}{128} = \frac{13}{64}.$$

This is an exact certificate *if* its premises have been established; it is not a measured Ashkin–Teller error. It exposes the substantial loss from returning through two analytic moves and many possible colors. Changing a finite trial or buffer can improve one term while making the holomorphic clearance and constants worse. There is no claim of an optimized physical algorithm.

6 Sharper fixed-buffer Gaussian shift budgets

Proposition 6.1 (Exact equal-covariance shift distance)**.** *Let μ be a centered Gaussian law with Cameron–Martin space $\mathcal{H}$, and let μ_u, μ_v be its translates by $u, v \in \mathcal{H}$. For $d = \|u - v\|_{\mathcal{H}}$ and standard normal distribution function Φ,*

$$\mathrm{TV}(\mu_u, \mu_v) = 2\Phi(d/2) - 1, \qquad \left\| \frac{\mathrm{d}\mu_u}{\mathrm{d}\mu} - \frac{\mathrm{d}\mu_v}{\mathrm{d}\mu} \right\|_1 = 4\Phi(d/2) - 2 \le \min\{2, \sqrt{2/\pi}\, d\}. \tag{18}$$

Proof. Translation reduces to $v = 0$. For $d > 0$ the likelihood is $\exp(dZ - d^2/2)$, where Z is a standard normal isonormal coordinate. It exceeds one exactly on $Z > d/2$. The difference of the shifted and unshifted probabilities of this event is $\Phi(d/2) - (1 - \Phi(d/2))$, proving total variation. The L^1 distance is twice this value. Integrating the normal density, bounded by $(2\pi)^{-1/2}$, gives the linear bound. The case $d = 0$ is immediate. The argument uses the Cameron–Martin coordinate, not pointwise values of a rough field. □

This classical identity sharpens source `eq:kernel-cm-continuity`, whose valid bound is $\min(2, d)$. For a $[0,1]$-valued cable test the expectation cost is total variation, not the L^1 distance. Suppose a common energy net has radius η both in the cable reference law and in its limiting reference law, and the error at every net point is at most e_{net}. The pure bounded-test comparison is then at most

$$e_{\mathrm{net}} + 2[2\Phi(\eta/2) - 1]. \tag{19}$$

For a product with a signed field functional of absolute value at most one, replace the second term by $4[2\Phi(\eta/2) - 1]$. Each side of the comparison is moved once to the net; the two costs are not silently collapsed to one.

If H_r is the independent harmonic correction in a Markov coupling between a domain and an inner exhaustion, and χ is the fixed cutoff equal to one near the observed buffer, conditioning on H_r gives

$$\mathrm{TV}(\text{two restricted field laws}) \le \mathbb{E}[2\Phi(\|\chi H_r\|_\nabla/2) - 1] \le \frac{\mathbb{E}\|\chi H_r\|_\nabla}{\sqrt{2\pi}}. \tag{20}$$

This is an upper bound after restriction, not necessarily equality: the cutoff contains information outside the observed buffer.

Domain and conditioning restrictions. The reference metric subgraph must be identical in every ambient approximation, with the same cable resistances and midpoint subdivisions. Conditioning on a perimeter clamps its entire cable trace to zero for the residual; deleting perimeter edges is not the same operation. The source's reverse-buffer proof additionally needs convergence of projection *norms*, not just covariance convergence for a finite list of marginal tests. The auxiliary cable mesh is separate from the current mesh. A fixed-buffer estimate such as (20) has no uniform content as that buffer approaches a rough boundary.

In the source no-crossing proof, canonical forcing is an identity on the original field space, whereas the upper bound conditions only on a retained stopping sigma-field. Adding canonical ancestor data to that conditioning can change the law. The local estimate (18) does not authorize that change and does not construct the common kernel by itself.

7 Compatible Ram coordinates and weak certificates

Fresh readings of the original private PDE treatment [8] give three useful rules: define the weak inverse and its domain before writing a Green kernel; impose the correct trace or gauge; and measure a residual in the whole dual space rather than in a solved finite system. These are the operative Ram connections here. They are not replacements for reflection positivity or for local-set geometry.

For real conforming basis functions $\psi_1, \dots, \psi_N \in H^1_0(P)$ and $u = \sum_j z_j \psi_j$, define

$$K_{ij} = \frac{1}{2\pi} \int_P \nabla\psi_i \cdot \nabla\psi_j, \qquad M_{ij} = \int_P \psi_i \psi_j, \qquad \|u\|_\nabla^2 = z^{\mathsf{T}} K z. \tag{21}$$

With complex functions use $\overline{\nabla\psi_i} \cdot \nabla\psi_j$ and $z^* K z$. If $z = Rw$, the physical forms are $R^* K R$ and $R^* M R$. A rectangular injective R selects a smaller space; it does not certify the omitted modes. In a source-coefficient representation the Green action is $K^{-1}M$ with the corresponding energy convention, whereas K^{-1} acts on the load vector. Confusing these changes the physical source measure.

Lemma 7.1 (Residual and shift-error composition)**.** *Let $V = H^1_0(P)$ with norm (3), and let $\mathfrak{a}$ be a bounded coercive form on V with constant $\alpha > 0$. If $u, \widetilde{u} \in V$ have the same weak boundary problem and*

$$\mathcal{R}(v) = \ell(v) - \mathfrak{a}(\widetilde{u}, v), \qquad \mathfrak{a}(u, v) = \ell(v) \quad (v \in V),$$

then $\|u - \widetilde{u}\|_\nabla \le \alpha^{-1} \|\mathcal{R}\|_{V'}$. Consequently a certified shift net error η and such a residual bound r/α can be combined in (18) using $d \le \eta + r/\alpha$.

Proof. For $e = u - \widetilde{u}$, coercivity gives $\alpha\|e\|_\nabla^2 \le |\mathcal{R}(e)| \le \|\mathcal{R}\|_{V'} \|e\|_\nabla$. Cancel if $e \neq 0$ and apply the triangle inequality to the two errors. □

The lemma is classical and is proved in the original private passage. It becomes useful only with a genuine upper bound on the dual norm. Finite residual samples give a lower bound, not that upper bound. A wrong boundary trace requires a lifting before the lemma applies. For a cutoff, the elementary product estimate

$$\|\chi e\|_\nabla \le \|\chi\|_\infty \|e\|_\nabla + \frac{\|\nabla\chi\|_\infty}{\sqrt{2\pi}} \|e\|_2$$

must also be charged. This shows explicitly how shrinking collars can worsen an apparently small approximation error.

7.1 An exact physical-norm fixture and an obstruction

On $P = (0,1)^2$, take $\psi = x(1-x)y(1-y)$. Polynomial integration gives $\int_P |\nabla\psi|^2 = 1/45$, so the scalar stiffness is $K = 1/(90\pi)$. The shift $u = z\psi$ has $d = |z|/\sqrt{90\pi}$ and exactly the variation distance (18). Rescaling the coordinate by a nonzero rational r changes the stiffness to r^2K and the coefficient to z/r, leaving the distance unchanged. This is a finite-dimensional calculation in the correct physical space, not a certificate for arbitrary harmonic shifts.

Indeed finitely many continuous linear measurements on $H_0^1(P)$ have a nonzero common kernel. To see this, choose more smooth, linearly independent compactly supported functions than measurements and solve the homogeneous finite system. A nonzero resulting function w has zero measured data but positive Dirichlet energy. The two shifts 0 and cw agree on those measurements for every c, while their full-field total variation tends to one as $|c| \to \infty$. Thus no finite Gram test alone provides the full local-kernel theorem.

8 What remains global, and what the local results preserve

The source's screening, Gaussian and local-kernel modules were read completely for this report. Selected spatial, stopped-field and final identification passages were also read. Their uses in the present report preserve the following distinctions.

Exact cuts, not arbitrary observations. Terminal data are measurable at the stated stop and exclude later interior revelations. A single lattice precursor can pinch into several continuum chambers. Their independence is established by a joint conditional Gaussian characteristic function with zero cross-component Green kernel, not by calling them geometrically different.

No unaccounted contact distribution. The source separately removes neighborhoods of the cut, truncates contributor counts and protected labels, and excludes collision measures by moment estimates. Harmonicity away from diagonals alone does not rule out a distribution supported on a diagonal. The discovered mean argument similarly controls invisible clusters and collapsing odd holes before asserting that no extra distribution remains on the explored set.

Same marginals need not give the same joint object. If h_1, h_2 are independent copies of a GFF, then (h_1, h_1) and (h_1, h_2) have identical individual covariances and different mixed covariance. Applying a cluster construction separately to independent fields likewise does not recover a prescribed same-field coupling. All-generation count bounds, canonical forcing and complete matching remain separate obligations, even after every estimate in this report has been verified.

Normalization and source corrections retained. The source's input-conventions appendix explicitly reconciles the column slope I_{col}/L with per-site free energy; an additional division by the number of columns is not permitted. Its Dirichlet energy has no one-half factor, while a Gaussian tail has the usual one-half in its exponent. It records finite factors before the final limits and elementary repairs in its quoted path and conformal-growth inputs. Those are *source-reported* corrections, not independently established errata to the external papers. No new source error is asserted here.

Gauge. The plane height is defined modulo an additive constant and is tested with exactly neutral coefficients, including after interpolation and reference averaging. In finite domains the zero-Dirichlet height has a fixed anchor. The Neumann compatibility and quotient discussion in the private original cannot be substituted for this Dirichlet problem. Nor can bare L^2 Fourier inversion justify a pointwise identity. The integrations in theorem 4.2 use compact smooth tests, finite positive mass and an explicit exponential majorant.

9 Evidence, literature and assessment

The accompanying read ledger records literal display intervals, their unread complements and hashes captured before and after reading. Structure searches are locators only. The complete external papers cited by the source were not audited. The frozen test program uses exact rational arithmetic for its algebraic fixtures; its counts are separate from any independent review. The tests check the formulas implemented, not reflection positivity for the model, uniform moment bounds, or the global scaling limit.

The public inventory has 7,845 lines in 13 texts; 5,056 distinct lines were literally displayed and read. The following compressed ledger specifies the scope without treating structural searches as proof reading.

File/module	Literal intervals
Main; bibliography	1–84; 1–283 (complete)
01; 02	1–103; 1–481 (complete)
03 spatial	1–315, 1200–1287, 1610–1715
04; 05; 06	1–462; 1–674; 1–1140 (complete)
07 local sets	229–465
08 local kernel	1–762 (complete)
09 identification	641–743, 1021–1066
10 conventions	1–172 (complete)
README	Unread; inventory and hash only
Private original [8]	53–195, 492–558: 210 distinct lines

In particular, the remaining circuit, collar, count, witness, thinness, canonical-geometry and complete-matching proofs are not claimed as fresh literal reads. The private core-pattern and two book-chapter leads were located but not read as proofs or used as support. Their unread status and every public complement are retained in the machine-readable ledger.

The owned test receipt reports 947 cases and 2,013 exact assertions. Of these, one case with six assertions checks the manuscript contract; the remaining 946 cases and 2,007 assertions are algebraic fixtures. They cover the joint edge table, rational ellipse identities, Cauchy-defect factorization, Laplace integration by parts, integer color budgets, sharp clipping, physical stiffness congruence and the explicit screening composition. No numerical Bessel, Gaussian-tail or analytic continuation test is being represented as an exact theorem certificate.

Current public status. The October 9, 2026 primary-source refresh confirms that Alcalde López–Heeney–Lis [2] constructs a GFF/critical Ising magnetization coupling. Alcalde López–Sepúlveda [3] studies the XOR-Ising excursion decomposition and states a broader Ashkin–Teller polarization scaling-limit conjecture in its abstract. These are related objects, not an external proof of every joint current-support statement in [1]. The DKLM abstract [4] states six-vertex GFF convergence for $c \in [\sqrt{3}, 2]$, with anisotropic weights treated using a suitable embedding. Its narrower height-convergence theorem is not a full-critical-line theorem. The source instead invokes other DKLM regularity, convergence-criterion and free-energy inputs after its own Gaussian-identification step. Those imported implications remain outside our independent proof claim. The local-set, first-passage and excursion literature [5, 6, 7] supplies the canonical objects and external convergence machinery. The present browsing scope is bibliographic/abstract checking, not a replacement audit of those deep theorems.

Advantages and costs. The explicit exponent makes a qualitative screening implication inspectable: one can see which clearance or translation worsens the budget and how a color tail turns a normalized error into an averaged one. The reflected spectral bound uses a positive defect already present in the source and keeps its out-of-band remainder visible. The Gaussian shift identity gives a strictly smaller local net cost than Pinsker's linear bound, in the correct Dirichlet

metric. The costs are real: multiple moves multiply exponents, very rare-pattern normalization needs multiplicative comparison, and fixed buffers cannot be allowed to collapse for free.

Feasible next steps. An actual mesh-rate result would need independently evaluated uniform reflection bounds, a rate for the angular-defect bands and their endcaps, and a quantitative short-witness/contiguity modulus. One could then minimize (5), (13) and (16) jointly over a *certified* set of layouts. A practical kernel calculation would also need reliable dual residual and boundary-lifting estimators, not only a well-conditioned finite Gram matrix. These are research tasks, not results claimed by the present report.

Contribution calibration. Applicability remains L3. The demonstrated contribution is a modest quantitative refinement of a specific proof interface: slit-ellipse smallness, reflected defect control and explicit finite screening composition, plus a classical sharp local shift constant. The analytic continuation, Cauchy comparison, Gaussian likelihood and weak-residual principles themselves are classical. Since their model-specific moduli have not been evaluated and the global identification has not been re-proved, C2 is appropriate; no claim of historical novelty or measured computational speedup is made.

References

[1] OpenAI, *The joint scaling limit of critical Ashkin–Teller currents*, September 25, 2026. Pinned build at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; public source directory. Primary objects of the present scoped study are modules 04–06 and 08.

[2] T. Alcalde López, L. Heeney and M. Lis, *The Ising magnetisation field and the Gaussian free field*, arXiv:2602.05886v1, February 5, 2026. Abstract and version record checked.

[3] T. Alcalde López and A. Sepúlveda, *Excursion decomposition of the XOR-Ising model*, arXiv:2602.06011v2, March 25, 2026. Abstract and version record checked.

[4] H. Duminil-Copin, K. K. Kozlowski, P. Lammers and I. Manolescu, *Gaussian free field convergence of the six-vertex model with* $-1 \leq \Delta \leq -\frac{1}{2}$, arXiv:2603.06268v1, March 6, 2026. Abstract and version record checked. The source's cited inputs are external; their proofs were not independently audited here.

[5] J. Aru, A. Sepúlveda and W. Werner, *On bounded-type thin local sets of the two-dimensional Gaussian free field*, arXiv:1603.03362v3. External local-set input, not independently audited here.

[6] J. Aru, T. Lupu and A. Sepúlveda, *The first passage sets of the 2D Gaussian free field: convergence and isomorphisms*, arXiv:1805.09204v1. External metric-graph input, not independently audited here.

[7] J. Aru, T. Lupu and A. Sepúlveda, *Excursion decomposition of the 2D continuum GFF*, arXiv:2304.03150v2. External excursion input, not independently audited here.

[8] M. Subbarao, private original PDE manuscript, `RamNonlin-IE-PDE-astra/TR8-lin-non-lin/TR8-lin-non-lin.tex`. Fresh literal reading: lines 53–195 and 492–558, covering weak Dirichlet inverses, trace/gauge distinctions, compatible coordinates and full-dual residual bounds. Confidential local source; not uploaded.